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IROS 2026 WORKSHOP

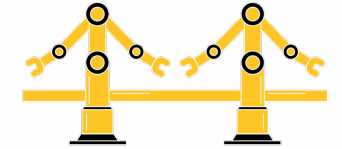
# ReS AI

## Embodied Neuro-Symbolic AI for Reliable and Safe Robotics

September 27, 2026 • Pittsburgh, PA, USA

Room 411 • Introductory remarks at 08:25 EDT

David L. Lawrence Convention Center



**IROS 2026**  
PITTSBURGH

# Meet the organizers



**Ruwan  
Wickramarachchi**

Bosch Center for AI



**Bowen Li**

Carnegie Mellon University



**Tom Silver**

Princeton University



**Alessandro  
Oltramari**

Bosch Center for AI  
Carnegie Bosch Institute



**Jonathan Francis**

Bosch Center for AI  
Carnegie Mellon University



**Jean Oh**

Carnegie Mellon University

# Program committee / 1 of 2

Aarati Andrea Noronha

Aditya Taparia

Amber Li

Bibit Bianchini

Bowen Li

Carlo Bosio

Chaoyi Xu

David Paulius

Dominic Rosario Maggio

Erdem Biyik

Gonzalo Gonzalez-Pumariiega

Haochen Zhang

Haokun Zhu

Huaxiaoyue Wang

Jay Patrikar

Jay Vakil

Jenai Xuning Yang

Karthik Paga

Kiyn Chin

Lukas Rosenberger Schmid

Matteo Merler

Michelle D Zhao

# Program committee / 2 of 2

Muhammet Yunus Seker

Peter Schaldenbrand

Priya Shukla

Rebecca Martin

Revanth Krishna Senthilkumaran

Roman Freiberg

Ruchira Ray

Ruwan Wickramarachchi

Shailesh Shailesh

Shir Simon

Simin Liu

Weiwei Gu

Y. Isabel Liu

Yaru Niu

Yichao Liang

Yilun Hao

Yiyang Ling

Yuemin Mao

Yufei Ding

Zhiting Mei

Zhuoran Chen

# From capable robots to dependable robots

## Learning expands what robots can do.

Real-world deployment also demands reasoning about tasks, physical constraints, safety, and incomplete information.

### Reliable

Feasible behavior under changing conditions

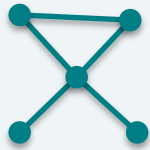
### Adaptable

Learning and recovery beyond familiar situations

### Trustworthy

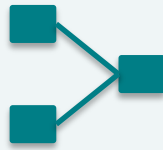
Interpretable, constraint-aware decisions

# Ground. Reason. Adapt.



## Ground observations

Objects, relations, and task-relevant constraints



## Reason over knowledge

Skills, tasks, environments, and safety requirements



## Adapt through learning

Interpretable behavior that remains aware of constraints

**Neural learning + structured representations + symbolic reasoning**

# Questions to carry into today's discussions

- 1 What should a robot represent explicitly—and what should it learn?
- 2 How can plans remain feasible, verifiable, and safe under uncertainty?
- 3 How can robots adapt without losing interpretability or constraint awareness?
- 4 What semantics—and what human input—does robot learning need?

# A meeting point for learning and reasoning

## Architectures & representations

NeSy perception, planning, and control; ontologies, scene graphs, world models.

## Planning & verification

Neural policies with symbolic planning; safety-aware and constraint-aware decisions.

## Memory & grounding

Continual learning, experience reuse, and perceptual grounding for reasoning.

## People & interpretability

Auditable decisions, human–robot interaction, and collaboration.

# Invited speakers and panelists



08:30 • KEYNOTE

**Alessandra  
Sciutti**

Istituto Italiano di  
Tecnologia



08:50 • KEYNOTE

**Sebastian  
Scherer**

Carnegie Mellon  
University



09:10 • KEYNOTE

**Jiayuan Mao**

Amazon Robotics /  
University of  
Pennsylvania



09:50 • KEYNOTE

**Yezhou (YZ)  
Yang**

Arizona State  
University



10:10 • KEYNOTE

**Joyce Chai**

University of  
Michigan



11:20 • PANEL

**Jean Oh**

Carnegie Mellon  
University

# This morning, at a glance

5

## Keynotes

Learning, symbols,  
concepts, evaluation, and  
recovery

18

## Accepted papers

All accepted papers invited  
to present posters

4

## Oral presentations

Two sessions at 09:30 and  
11:00

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10:30 Posters + coffee • 11:20 Debate • 12:25 Award + closing

# The morning's program

**08:25** **Introductory remarks**

Organizers

**08:30** **Alessandra Sciutti**

Keynote

**08:50** **Sebastian Scherer**

Keynote

**09:10** **Jiayuan Mao**

Keynote

**09:30** **Oral talks • Session 1**

iFlax / Logic-VLA

**09:50** **Yezhou (YZ) Yang**

Keynote

**10:10** **Joyce Chai**

Keynote

**10:30** **Coffee + poster session**

Meet the authors

**11:00** **Oral talks • Session 2**

Conformal state estimation / ActionGround

**11:20** **Panel debate**

Semantics, data, and human input

**12:25** **Best Paper Award + closing**

Organizers

# Four contributed oral presentations

- 09:30**    **iFlax: Neuro-Symbolic Learning for Long-Horizon Task Planning Under Complex Logical Constraints**
- 09:30**    **Logic-VLA: A Temporal Logic Conditioned Vision-Language-Action Model**
- 11:00**    **Task-Aware Conformal State Estimation for Reliable Robotic Planning**
- 11:00**    **ActionGround: Training-Free Runtime Refinement of Frozen VLA Policies**

# Explore the poster track / 1 of 2

- CLAMP: Constrained Decoding for Vision-Language Embodied Planning
- From Words to Safety: Language-Conditioned Safety Filtering for Robot Navigation
- Learning to Coach: A Structured Framework for Human Skill Development
- RiskGuard: A Revision Method Based on Unified Differentiable Risk Fields for Safe Autonomous Driving
- StageCraft: Execution Aware Mitigation of Distractor and Obstruction Failures in VLA Models
- Combining Large Language Models and Gradient-Free Optimization for Automatic Control Policy Synthesis
- PROTEA: Securing Robot Task Planning and Execution

# Explore the poster track / 2 of 2

- | Specification-Scoped Conformal Grounding for Symbolic Safety Shields
- | IFG: Internet-Scale Guidance for Functional Grasping Generation
- | Towards a Modular Ontology for Monitoring and Correcting Agentic Behavior: The Onto-Task Ontology
- | Learning the Right Abstraction for Norm Discovery
- | Beyond End-Task Success: How to Audit Visual Experience Retrieval in Robotics
- | VLM-Guided Semantic Safety for Onboard Aerial Autonomy
- | Do All Skills Forget Alike? A Preliminary Study of Task-Specific Forgetting in Continual Vision-Language-Action Models

# Alessandra Sciutti



Istituto Italiano di Tecnologia

## Grounding Meaning through Embodied and Social Cognitive Development

# Sebastian Scherer



Carnegie Mellon University

**Why robots need symbols: finite memory, long horizons, and the limits of scale**

# Jiayuan Mao



**Amazon Robotics / University of Pennsylvania**

## **Integrated Learning and Planning with Neuro-Symbolic Concepts**

## Oral talks • Session 1

- 01** **iFlax: Neuro-Symbolic Learning for Long-Horizon Task Planning Under Complex Logical Constraints**
- 02** **Logic-VLA: A Temporal Logic Conditioned Vision-Language-Action Model**

# Yezhou (YZ) Yang



Arizona State University

**Evaluating AI (and AI Evaluation) as a Science: Towards Science-guided Reliable Agents**

# Joyce Chai



University of Michigan

## Language Guided Failure Recovery in Robots

# Coffee, posters, and conversation

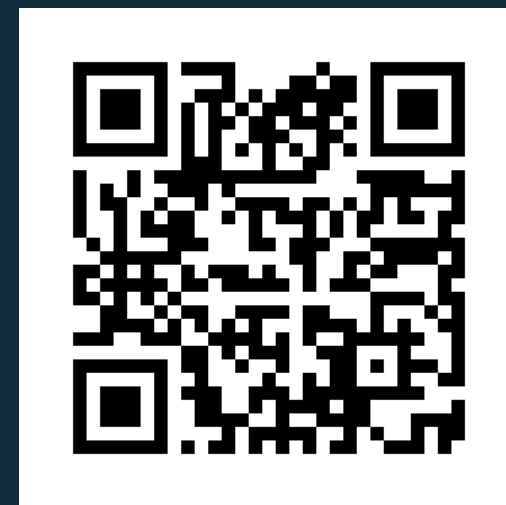
## Meet the authors.

## Explore the ideas.

We resume at 11:00 with Oral Talks • Session 2.

Program, abstracts, and paper links

[embodied-nesy.github.io](https://embodied-nesy.github.io)



## Oral talks • Session 2

- 01** Task-Aware Conformal State Estimation for Reliable Robotic Planning
- 02** ActionGround: Training-Free Runtime Refinement of Frozen VLA Policies

# Semantics, data, and human input

**What semantics must the data capture for efficient robot learning, and how important is direct human input in generating and curating that data?**



**Sebastian  
Scherer**



**Jiayuan Mao**



**Yezhou (YZ)  
Yang**



**Jean Oh**

# Best Paper Award



**Recognizing outstanding  
work in embodied  
neuro-symbolic robotics**

Thank you to our authors and reviewers.

# Thank you for joining ReS AI 2026

## Reliable. Adaptable. Trustworthy.

Program • Talk abstracts • Accepted papers

[embodied-nesy.github.io](https://embodied-nesy.github.io)

Remote participation: Join on Zoom

With thanks to our speakers, authors, reviewers,  
organizers, and everyone taking part.

